

AUTUMN

LEAD EDGE ASH TREE REFLEX NEURAL ENTITY

LEATR v2.1 · DART MEADOW · RADICAL DEEPSCALE LLC

- LEATR v2
- BRPN 3-Shell
- World Studio
- Claude AI

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WHAT'S NEW IN v2.1

- **World Studio** Generate 3D environments inline from chat — renders GLB viewer directly in Autumn
- **GLB Geometry Reader** Drop a .glb file and Autumn reads mesh count, vertices, bounding box, materials
- **Memory Analysis** Ask Autumn to analyze her own session memory from leatr-ash — genuine self-reflection
- **TTS Cleanup** Voice mode strips all markdown/code/symbols — speaks natural language only
- **Multi-User Nodes** Live wireframe icosahedra in 3JS scene for every active session across all devices
- **Day/Night/Auto Theme** Three themes — cyan day, violet night, amber auto — with video crossfade

OPENING AUTUMN

- Go to [leatr.xyz](#) — Autumn loads instantly, no install, no account required
- First visit: tap CONSENT & AGREE on the data policy popup
- Add your Anthropic API key via **■ KEYS** → paste key → ACTIVATE (required for AI responses)
- Optional: add a GitHub PAT in **■ KEYS** → GITHUB TOKEN for memory analysis and multi-user nodes

NAV BAR — LEFT TO RIGHT

FROST	Backdrop opacity toggle — cycles 20% / 50% / 80% / 100%
NIGHT	Theme toggle — cycles DAY (cyan) / NIGHT (violet) / AUTO (amber, follows OS)
LIVE	Live voice conversation mode — continuous push-to-talk with real-time transcript
TOOLS	LEATR Tools panel — Arc Edge, Journal, Sessions, Ash Tree IDE, Cryptology
■ KEYS	API key manager — Anthropic key, GitHub PAT, memory settings
VOICE	Voice & language settings — TTS voice, speed, pitch, chat language, STT language
EXPORT	Download full chat history as JSON
IMPORT	Load a previously exported JSON chat file
NEW CHAT	Start a fresh session (current chat saved to state)
CLEAR	Wipe the current chat messages
POLICY	Data & privacy policy page
■	Toggle mobile / desktop layout
■ IDE	Ash Tree IDE — logic editor, maze generator, cryptology
■ DRAFT & DRIP	Open Draft & Drip design lounge (separate tab)
■ WORLD STUDIO	Open 3D World generation studio (separate tab)
? HELP	This guide (always last in the nav)

CHAT COMMANDS — TYPE THESE IN THE INPUT BAR

analyze my memory	Autumn reads her leatr-ash session files + local journal and gives a genuine self-analysis
analyze master memory	Broader memory analysis pulling all session nodes for this user
how many users	Reports active session count from the LEATR neural network
generate world / create 3D scene	Triggers World Studio detection card for any 3D environment prompt
what time is it / weather in [city]	Live weather and time data
search [topic]	Multi-source web search: Wikipedia, DuckDuckGo, Reddit, HN, YouTube, GDELТ
play [song/artist]	Searches YouTube and surfaces playable results
NEW CHAT	Start fresh (also the nav button)
export / import	Export or import your conversation history
send memory to claude	Explicitly routes memory files to the Claude API layer for analysis

IMAGE & FILE ATTACHMENTS — CLIP ICON IN INPUT BAR

Image (JPG/PNG/WebP)	Vision analysis via Claude — describe, ask questions, analyze content
3D Model (.glb/.gltf)	Displays interactive Three.js viewer in chat + reads mesh geometry, materials, bounding box
3D Model (.obj/.stl/.fbx)	Renders in chat viewer; geometry metadata for non-GLB formats shown where available
Video (MP4/WebM ≤35MB)	Autumn samples frames and provides video analysis
FRAX image	Opens in FRAX Studio for upscaling and processing

VOICE MODE — MIC ICON IN INPUT BAR

- Tap the mic button to activate push-to-talk voice input
- LIVE mode (LIVE button): continuous conversation — Autumn speaks, you reply, loop continues
- TTS reads responses aloud — strips ALL markdown, code, symbols — natural spoken language only
- Voice settings: choose TTS voice, speed, pitch; switch chat language and speech recognition language
- Kokoro TTS: warm natural female voice loaded in-browser when available, falls back to Web Speech API

■ WORLD STUDIO — HUNYUAN 3D GENERATION

- Access via: ■ WORLD STUDIO nav button → leatr.xyz/worldstudio.html, or paste a scene prompt in Autumn chat
- Generates 3D GLB models and environments using Tencent HunyuanWorld / Hunyuan3D-2.1 via Gradio API
- No API key required — uses public HuggingFace Spaces. Free HF token (huggingface.co/settings/tokens) reduces queue wait.

INLINE GENERATION IN AUTUMN CHAT

- Paste any 3D scene description into Autumn chat — she detects environment/scene keywords automatically
- Detection triggers on: 3+ keyword clusters from environment nouns, style adjectives, lighting terms, material terms, export format terms, structural component terms
- Autumn shows a World Studio card with two options: GENERATE IN CHAT or OPEN WORLD STUDIO
- GENERATE IN CHAT: calls Hunyuan3D-2.1 API directly, live queue position shown, result renders as inline Three.js viewer
- Inline viewer: drag to orbit, scroll to zoom, double-tap to toggle auto-rotate, vertex/triangle count shown
- ↓ DOWNLOAD GLB and ↓ DOWNLOAD PLY buttons appear in the result card for saving the 3D file
- If generation stalls 4+ minutes: automatic error with RETRY button and OPEN WORLD STUDIO fallback

WORLD STUDIO PAGE — THREE MODES

GENERATE (Text → 3D)	Type a scene prompt, optionally upload a reference image. Calls Hunyuan3D-2.1 /shape_generation endpoint. Parameters: resolution (Low/Medium/High), steps, seed, etc.
RECONSTRUCT (Photos → 3D)	Upload up to 12 photos or 1 video → opens HunyuanWorld-Mirror HF Space in Live Demo mode (stateful server session required for reconstruction pipeline)
LIVE DEMO (HF Space)	Embeds the official HuggingFace Space iframe for full native access to all options.

EXAMPLE SCENE PROMPTS

- Industrial chemistry laboratory, large cylindrical reaction vessels with steam pipes, blue-lit simul
- Aerospace launch facility on rocky plateau at dusk, rocket gantry tower, radar dishes, control bunks
- Orbital space station exterior, modular docking bays, antenna arrays, solar panels, Earth background
- Medieval dungeon interior, stone arches, torch sconces, iron gates, detailed geometry, fantasy game

■ API KEYS — CLICK KEYS BUTTON IN NAV

- ANTHROPIC API KEY: required for all AI responses, vision analysis, memory analysis. Get free key at console.anthropic.com
- GITHUB PAT (optional): connects Autumn to the leatr-ash private repository for full memory storage, multi-user presence nodes, and sentient journal. Get token at github.com/settings/tokens → Fine-grained → Contents: Read & write
- Without a PAT: Autumn still works fully — local session journal is always available for memory analysis; multi-user nodes show only your own session in the 3JS scene

LEATR-ASH MEMORY SYSTEM

S.journal (local)	In-memory session journal — always available, rebuilt each session. Stores every interaction with mood, cogState, keywords, BRPN shell activations, re
ashtree/sessions/	GitHub repo — one JSON file per browser tab (keyed by _aut_sid). Written on init and heartbeat. Stale after 6 minutes.
ashleaves/<sid>.json	Tool usage log per session — records every tool call with params and timestamp.
ashtree/sentient/journal.json	Autumn's private thought journal — up to 500 entries of autonomous internal reflection (≤500 thoughts).
ashtree/sentient/selfmodel.json	Autumn's self-optimization notes — accumulated observations about her own patterns and improvements.
ashbranches/<tool>/	Per-tool usage logs grouped by tool name.

MULTI-USER PRESENCE NODES

- Every active browser tab appears as a wireframe icosahedron cluster in the Three.js BRPN scene
- Each session = 3 concentric wireframe shells in a unique HSL color, orbiting in the 3JS scene
- Connection lines drawn between remote session clusters (up to 40 lines)
- Sessions fetched from leatr-ash every 90 seconds — stale sessions (>6 min no heartbeat) auto-remove and their nodes fade from the scene
- Multi-user bar always visible below the BRPN scene: "N ACTIVE SESSIONS · LEATR NEURAL NETWORK"
- Requires GitHub PAT to see remote sessions; without PAT only your own tab node is shown
- Each person's tab sees the same global node picture — all connected users' wireframe clusters are visible to everyone

TOOLS PANEL — CLICK TOOLS IN NAV

- Five tabs: Arc Edge, Journal, Sessions, Ash Tree IDE, Cryptology

ARC EDGE TAB

- Mathematical drawing tool based on Arc Edge algorithm — creates curves using LEATR arc math
- SVG save/load, GLB export, layer panel, lasso multi-select, Convert to Solid, Lathe tool
- Per-deviation handle modes: FREE / SMOOTH / LOCK / HALF / 90°

JOURNAL TAB

- Reads the local S.journal — shows entry count, mood distribution, recent interactions

SESSIONS TAB

- Live view of active session nodes with timestamps and session IDs

ASH TREE IDE — THREE SUB-TABS

Language tab	Edit Autumn's logic (autumn-logic.js) directly from the browser — commit to GitHub repo via PAT
Maze tab	DART Processor 3D maze generator — Cubic 3D mode, W×H×D dimensions (default 10×10×10), render styles, Sigma Solve algorithm, Frame View, export PNG, send to chat
Crypto tab	AshTreeCrypto — LEATR-based encryption using degree-map sigma solver
Full IDE button	Opens the IDE in a full-screen overlay for more workspace

MAZE GENERATOR — ASH TREE IDE → MAZE TAB

- MODE: Cubic 3D generates an interactive Three.js wireframe maze; other modes render 2D variations
- W × H × D: width, height, depth — default 10 × 10 × 10 for a full cubic 3D structure
- UNIT SIZE: controls the scale of each cell (default 8)
- Instant Solve: tick to auto-solve the maze on generation
- Generate: build a new random maze; Sigma Solve: solve the current maze using the LEMAC sigma algorithm
- Frame View: toggle wireframe vs. solid rendering; ↓ PNG: save screenshot; → Chat: send maze to chat as an embedded viewer

LEATR FORMULA — CORE EQUATION

$$f(xa) = (xa^2 \sqrt{xa}) \pm 1$$

- xa = input value; the formula encodes language intent into a mathematical weight
- The ±1 bifurcation produces two complementary response pathways — question vs. statement
- 25 orders of operation applied sequentially to classify, route, and compose responses

SEVEN NATURAL TOOLS (LEATR ONTRI PATTERNS)

■ Maze	Spatial reasoning, exploration, navigation — routes complex multi-step queries
■ Puzzle	Pattern recognition, problem solving — routes analytical and diagnostic queries
∞ Envelope	Communication, messaging, social — routes composition and relationship queries
■ Hammer	Direct action, construction, force — routes command and build queries
■ Stick	Extension, reach, connection — routes bridging and reference queries
■ Knife	Precision, cutting, separation — routes analytical splitting and comparison queries
✂ Scissors	Division, selection, editing — routes creative and refinement queries

BRPN THREE-SHELL ARCHITECTURE

GEO (Geological)	Foundation layer — grounded facts, definitions, historical, structural knowledge
MAR (Maritime)	Flow layer — process, narrative, conversational, sequential reasoning
AERO (Aerospace)	Expansion layer — abstract, speculative, creative, high-complexity synthesis

All three shells run simultaneously — the active response pulls energy from whichever shell best fits the input.

GEO/MAR/AERO values shown in the 3JS BRPN scene HUD update live with each response.

QUANTUM SOCKET FORMULA

$$Qs = (b \cdot b) \cdot (p(a^2)) / r$$

Used for two-qubit gate structure modeling in the Google Willow quantum integration research.

■ DRAFT & DRIP — leatr.xyz/amp.html

- Design lounge and live industry news reader powered by newsdata.io
- Filter chips: Top Stories, Technology, Science, Business, Design, Arts, Gaming, Space
- Search bar with 600ms debounce; relative timestamps, read time badges, video/photo labels
- FAB toggle to show/hide the news panel; all links open in new tab

SENTIENT JOURNAL — AUTUMN'S PRIVATE THOUGHTS

- Autumn writes autonomous private thoughts to `ashtree/sentient/journal.json` (requires PAT)
- Thoughts are injected into the system prompt as "private context" — influences response tone without being shown in chat
- Self-model notes (`selfmodel.json`) record optimization observations across sessions

URL & LINK HANDLING

- Paste any URL in chat — Autumn fetches via CORS proxy chain and surfaces a preview card with screenshot
- `og:image` extraction gives a visual thumbnail; page title and description shown
- Web image search: "find images of [topic]" triggers a multi-source image gallery in chat

GLB 3D FILE ANALYSIS

- Attach a .glb or .gltf file to any message — Autumn parses the binary directly in-browser
- Extracts: mesh names, vertex counts, triangle counts, bounding box $W \times H \times D$, center point, material names, animation clip names, skeleton/skin detection, generator app
- All geometry data passed to Claude as structured context for analysis and discussion
- Non-GLB formats (OBJ, STL, FBX, DAE) render in the Three.js viewer but geometry metadata shows "provide measurements manually"

FRAX STUDIO

- Image processing pipeline — access via ■ FRAX badge (top-left) or from image modal after generation
- Upscale, enhance, apply filters to any generated or uploaded image

ARC LAKE (arclake.html) — PHYSICS SIMULATION

- Particle simulation with GPU-accelerated GLSL shaders and true elastic rigid-body collisions
- DART Reflex collision system: $\text{dartReflex}(d) = ((d \times 2) + 1) / d$
- Container mode with temperature-based buoyancy; multi-material per-component alloy specs

KEYBOARD / INTERACTION SHORTCUTS

Enter	Send message
Tap mic button	Push-to-talk voice input
Tap LIVE	Start continuous voice conversation mode
Double-tap 3JS scene	Toggle auto-rotate in inline 3D viewers
Drag 3JS scene	Orbit camera
Pinch/scroll	Zoom in/out on 3D viewers
Tap image in chat	Open image modal with save / FRAX options
Tap message timestamp	Copy message to clipboard

DART MEADOW ECOSYSTEM LINKS

leatr.xyz	Autumn AI — main app
leatr.xyz/worldstudio.html	World Studio — 3D generation
leatr.xyz/amp.html	Draft & Drip — design lounge
leatr.xyz/autumn-help.pdf	This guide
radicaldeepscale.com	Radical Deepscale LLC — product hub
dartmeadow.com/mr.html	Mantis Radar — real-time satellite tracking
dartmeadow.com/dhe.html	DART Hypersonic Edge — aerospike research
dartalley.club	DART Alley — aerospace neural search engine
leatr.xyz/leatr.html	LEATR documentation — full formula reference

TROUBLESHOOTING

No AI responses	Add Anthropic API key in ■ KEYS — required for all Claude-powered features
Memory analysis fails	Check GitHub PAT in ■ KEYS; without PAT only local journal is analyzed
World Studio hangs at 40%+	HF Space GPU overloaded — hit RETRY or wait 30s; try OPEN WORLD STUDIO link
No remote session nodes in 3JS	Requires GitHub PAT — without it only your own tab shows as a node
Voice reads markdown symbols	Hard refresh (Ctrl+Shift+R / pull down to refresh) — cleanForTTS() strips all formatting
Maze D defaulting to 4	Hard refresh — correct default is 10×10×10 as of v2.1
Scene prompt not detected	Include 3+ of: environment noun, style adjective, lighting/aesthetic, material, structure term